

Algebra Reference Sheet

MATH-310 — Anna Lachowska

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Abstract. This sheet summarizes the main techniques and theorems needed to solve algebra exams. Good luck!

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Part I: Group Theory

1. Basic Definitions

Group. Set G with operation satisfying: closure, associativity, identity e , inverses.

Order. $|G|$ = number of elements. $o(g)$ = smallest $n > 0$ with $g^n = e$.

Subgroup. $H \leq G$: nonempty subset closed under operation and inverses.

Normal subgroup. $H \triangleleft G$ iff $gHg^{-1} = H$ for all $g \in G$.

- Index 2 subgroups are always normal.
- To check: verify $gHg^{-1} = H$ only for generators of G .

2. Group Actions & Subgroups

Orbit. $\text{Orb}_G(x) = G \cdot x = \{g \cdot x : g \in G\}$

Stabilizer. $\text{Stab}_G(x) = G_x = \{g \in G : g \cdot x = x\}$

Orbit-Stabilizer. $|G| = |\text{Orb}(x)| \cdot |\text{Stab}(x)|$

Center. $Z(G) = \{g \in G : gx = xg \ \forall x \in G\}$ (always $\triangleleft G$; for p -groups: $\neq \{e\}$)

Centralizer. $C_G(x) = \{h \in G : hx = xh\}$

Note: $Z(G) \subseteq C_G(x)$ for all x . Also $C_G(x) = \text{Stab}(x)$ under conjugation action.

Class equation. $|G| = |Z(G)| + \sum_i [G : C_G(x_i)]$

where x_i are representatives of non-trivial conjugacy classes.

3. Key Theorems

Lagrange. $H \leq G$ finite $\Rightarrow |H|$ divides $|G|$.

Cauchy. p prime, $p \mid |G| \Rightarrow \exists g \in G$ with $o(g) = p$.

$|G| = p$ **prime.** G is cyclic (hence abelian).

$|G| = p^2$. G is abelian.

Why? Class equation $\Rightarrow |Z(G)| \geq p$. If $|Z(G)| = p$, then $G/Z(G)$ cyclic $\Rightarrow G$ abelian. So $Z(G) = G$.

4. Homomorphisms

Group hom. $\phi : G \rightarrow H$ with $\phi(ab) = \phi(a)\phi(b)$.

Ring hom. $\phi : R \rightarrow S$ with $\phi(a+b) = \phi(a) + \phi(b)$, $\phi(ab) = \phi(a)\phi(b)$, $\phi(1) = 1$.

Isomorphism. Bijective homomorphism.

Kernel. $\ker(\phi) = \phi^{-1}(e)$. For groups: $\ker \triangleleft G$. For rings: $\ker$ is an ideal.

First Isomorphism Thm. $G/\ker(\phi) \cong \text{Im}(\phi)$.

Hom existence:

- Group: $\text{Hom}(C_m, C_n) \neq 0 \iff \gcd(m, n) > 1$.
(C_n has element of order $d \mid m$ iff $d \mid \gcd(m, n)$)
- Ring: $\mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ (unital) exists iff $n \mid m$.

5. Symmetric Groups

S_n = bijections of $\{1, \dots, n\}$, $|S_n| = n!$

Cycle notation. $(a_1 \dots a_k)$: $a_i \mapsto a_{i+1}$, $a_k \mapsto a_1$.

Order. Disjoint cycles: lcm of lengths.

Parity. k -cycle = $(k-1)$ transpositions. Even iff k odd.

A_n . Even permutations. $|A_n| = n!/2$. Contains: e , 3-cycles, double transpositions.

Conjugation. $\sigma(i_1 \dots i_k)\sigma^{-1} = (\sigma(i_1) \dots \sigma(i_k))$.

Preserves cycle type. Conjugacy classes = cycle types.

Non-disjoint products:

- $(1\ 2 \dots k)(i\ m)$, $i \leq k < m$: becomes $(k+1)$ -cycle.
- $(1\ 2 \dots k)(i\ j)$, $i < j \leq k$: splits into two cycles.

6. Finite Abelian Groups

Structure Thm. $G \cong \mathbb{Z}_{p_1^{e_1}} \times \dots \times \mathbb{Z}_{p_k^{e_k}}$ (prime powers).

Counting. # abelian groups of order $n = p_1^{a_1} \dots p_r^{a_r}$ is $\prod_i p(a_i)$, where $p(a)$ = partitions of a .

Partitions of small n : $p(1) = 1$, $p(2) = 2$, $p(3) = 3$, $p(4) = 5$, $p(5) = 7$.

Invariant factors. $d_1 \mid d_2 \mid \dots \mid d_r$ with $G \cong \mathbb{Z}_{d_1} \times \dots \times \mathbb{Z}_{d_r}$.

Order existence in finite abelian group:

- Max element order = d_r (largest invariant factor).
- Element of order n exists iff $n \mid d_r$.

In cyclic group C_m : Element of order n exists iff $n \mid m$.

(Cyclic group of order m has exactly one subgroup of order d for each $d \mid m$.)

Order in products. For $(a, b) \in C_m \times C_n$: $\text{ord}(a, b) = \text{lcm}(\text{ord}(a), \text{ord}(b))$.

Max order in $C_m \times C_n$ is $\text{lcm}(m, n)$. Generalizes: max order in $C_{n_1} \times \dots \times C_{n_k}$ is $\text{lcm}(n_1, \dots, n_k)$.

Cyclic test. $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$ iff $\gcd(m, n) = 1$.

Why? $\text{lcm}(m, n) = mn$ iff $\gcd(m, n) = 1$. Only then $\exists$ element of order mn .

Part II: Rings & Fields

7. Ring Hierarchy

Field $\subset$ Euclidean Domain $\subset$ PID $\subset$ Integral Domain $\subset$ Ring

Integral domain. Commutative, $1 \neq 0$, no zero divisors.

Zero divisor. $a \neq 0$ with $ab = 0$ for some $b \neq 0$.

Unit. Has multiplicative inverse. Units are never zero divisors.

Irreducible. $c \neq 0$, not unit, $c = ab \Rightarrow a$ or b unit.

8. Fields & Characteristic

Field. Every nonzero element is a unit.

Finite fields. $|\mathbb{F}_q| = p^n$ for prime p . Unique up to isomorphism.

$\mathbb{F}_q$ exists iff $q = p^n$. (No $\mathbb{F}_{12}$, $\mathbb{F}_{18}$.)

Unit group. $\mathbb{F}_q^\times$ is **cyclic** of order $q-1$.

Characteristic. Let $\tau : \mathbb{Z} \rightarrow R$ be the unique ring hom ($\tau(1) = 1_R$). $\text{char}(R) = n$ where $\ker(\tau) = n\mathbb{Z}$. Equivalently: $\text{char}(R) = \min\{n > 0 : n \cdot 1_R = 0\}$, or 0 if none.

$$\begin{aligned} \text{char}(\mathbb{Z}/n\mathbb{Z}) &= n & \text{char}(\mathbb{F}_p) &= p & \text{char}(\mathbb{Q}) &= 0 \\ \text{char}(A \times B) &= \text{lcm}(\text{char } A, \text{char } B) \end{aligned}$$

9. Ideals & Quotients

Ideal operations (in PID):

- $(a)(b) = (ab)$
- $(a) \cap (b) = (\text{lcm}(a, b))$
- $(a) + (b) = (\gcd(a, b))$

Ex: $(6\mathbb{Z}) \cap (15\mathbb{Z}) \cap (10\mathbb{Z}) = (30\mathbb{Z})$.

Quotient ring. R/I : elements are cosets $[a] = a + I$.

In $F[X]/(p)$:

- Size: $|F|^{\deg p}$. Elements: polynomials of degree $< \deg p$.
- Field iff p irreducible. (Irreducible $\Rightarrow (p)$ maximal.)
- $\bar{f}$ unit iff $\gcd(f, p) = 1$. $\bar{f}$ zero divisor iff $\gcd(f, p) \neq 1$.

10. Irreducibility Tests

Degree ≤ 3 . Irreducible iff no roots in F .

Rational root test. $f \in \mathbb{Z}[X]$, root $\frac{r}{s}$ reduced $\Rightarrow r \mid a_0$, $s \mid a_n$.

Eisenstein. Prime p : $p \nmid a_n$, $p \mid a_i$ for $i < n$, $p^2 \nmid a_0 \Rightarrow$ irred. over $\mathbb{Q}$.

Reduction mod p . $\bar{f} \in \mathbb{F}_p[X]$ irred. with same degree $\Rightarrow f$ irred. over $\mathbb{Q}$.

11. Euler & Number Theory

Euler totient. $\varphi(n)$ = count of $1 \leq a \leq n$ with $\gcd(a, n) = 1$.

Formulas:

- $\varphi(p^k) = p^{k-1}(p-1)$
- $\varphi(mn) = \varphi(m)\varphi(n)$ when $\gcd(m, n) = 1$
- $\varphi(n) = n \prod_{p \mid n} (1 - 1/p)$

Product of primes:

- Distinct: $\varphi(pqr) = (p-1)(q-1)(r-1)$
- Two equal: $\varphi(p^2q) = p(p-1)(q-1)$
- All equal: $\varphi(p^3) = p^2(p-1)$

Units. $(\mathbb{Z}/n\mathbb{Z})^\times$ has order $\varphi(n)$.

Euler's Thm. $\gcd(a, n) = 1 \Rightarrow a^{\varphi(n)} \equiv 1 \pmod{n}$.

Fermat. p prime $\Rightarrow a^{p-1} \equiv 1 \pmod{p}$ (if $p \nmid a$).

12. Chinese Remainder Theorem

CRT isomorphism. $\gcd(m, n) = 1 \Rightarrow \mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.

Converse. $\gcd(m, n) > 1 \Rightarrow \mathbb{Z}/mn\mathbb{Z} \not\cong \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$.

Why? $\text{char}(\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}) = \text{lcm}(m, n) \neq mn$.

Unit count. $|(\mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z})^\times| = \varphi(m)\varphi(n)$.

Solving systems. $x \equiv a_i \pmod{n_i}$ with coprime n_i : unique solution mod $N = \prod n_i$.

Non-coprime. Solvable iff $a_i \equiv a_j \pmod{\gcd(n_i, n_j)}$ for all i, j .

13. Ring Isomorphism Invariants

To show $A \not\cong B$, find an invariant that differs:

- Characteristic: $\text{char}(A) \neq \text{char}(B)$
- Size: $|A| \neq |B|$
- Unit count: $|A^\times| \neq |B^\times|$
- Zero divisor count

Ex: $\text{char}(\mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/12\mathbb{Z}) = 36 \neq 108 = \text{char}(\mathbb{Z}/108\mathbb{Z})$.

Problem-Solving Strategies

Groups & Permutations

Prove G is abelian

If $|G| = p^2$: Always abelian (class equation argument).

If $|G| = 4$: Either cyclic, or all non- e have order 2.

In latter case: $(xy)^2 = e \Rightarrow xy = y^{-1}x^{-1} = yx$.

General: Cyclic $\Rightarrow$ abelian. Or check $xy = yx$ for generators.

Classify abelian groups of order n

- Factor $n = p_1^{a_1} \cdots p_r^{a_r}$.
- For each $p_i^{a_i}$, list partitions of a_i .
- Total count = $\prod_i p(a_i)$.
- Write groups using elementary or invariant factors.

For smallest max order: use partition with smallest largest part.

Show no subgroup of order k exists

For normal subgroups: must be union of conjugacy classes.

- List conjugacy class sizes (include 1 for identity).
- Check if any subset sums to k .
- No valid subset $\Rightarrow$ no such subgroup.

Ex: A_4 classes: 1, 3, 4, 4. No subset sums to 6.

Simplify non-disjoint cycle products

$(12 \cdots k) \cdot (ij)$: Apply right-to-left. Trace each element.

Case 1: $i \leq k < j$ — absorbs j into cycle, gives $(k+1)$ -cycle.

Ex: $(1234567)(38) = (12384567)$.

Case 2: $i < j \leq k$ — splits cycle into two.

Case 3: $k < i < j$ — disjoint, order = $\text{lcm}(k, 2)$.

Show $Z(G)$ and $C_G(x)$ are subgroups

Check subgroup criteria: (1) contains e , (2) closed under product, (3) closed under inverse.

For $Z(G)$: if $z_1g = gz_1$ and $z_2g = gz_2$ for all g , then $(z_1z_2)g = g(z_1z_2)$.

Note: $C_G(x)$ = stabilizer of x under conjugation action.

Rings & Polynomials

Analyze quotient ring $F[X]/(p)$

- Size:** $|F|^{\deg p}$
- Elements:** Polynomials of degree $< \deg p$
- Field?** Yes iff p irreducible
- Units:** $\bar{f}$ unit iff $\gcd(f, p) = 1$
- Zero divisors:** $\bar{f}$ ZD iff $\gcd(f, p) \neq 1$

If field: Unit group is cyclic of order $|F|^{\deg p} - 1$.

Ex: $\mathbb{F}_3[X]/(X^2 - X - 1)$ is field with 9 elements. Units $\cong C_8$.

Find gcd in $\mathbb{F}_p[X]$ + Extended Euclidean

- Divide: $f = qg + r$, $\deg r < \deg g$. Reduce coeffs mod p .
- Replace: $f \leftarrow g$, $g \leftarrow r$. Repeat until $r = 0$.
- Last nonzero remainder = gcd. Make monic.

Extended: Track r, s with $\gcd = fr + gs$ (Bézout).

In $\mathbb{F}_5$: $-1 \equiv 4$, $-2 \equiv 3$, $2^{-1} \equiv 3$, $3^{-1} \equiv 2$.

Test irreducibility

Degree ≤ 3 : No roots $\Rightarrow$ irreducible.

Over $\mathbb{Q}$:

- Rational root test: r/s root $\Rightarrow r \mid a_0$, $s \mid a_n$.
- Eisenstein with prime p .
- Reduction mod p (same degree, irreducible mod p).

Over $\mathbb{F}_p$: Test all roots $0, 1, \dots, p-1$.

Number Theory

Prove $n \mid a$ using modular arithmetic

- Factor $n = \prod p_i^{k_i}$.
- Show $a \equiv 0 \pmod{p_i^{k_i}}$ for each prime power.
- By CRT, this implies $n \mid a$.

Use Fermat/Euler to reduce exponents.

Ex: $30 \mid (p^{88} - p^{16})$ for $p \geq 7$? Check mod 2, 3, 5.

Solve CRT system

Given $x \equiv a_i \pmod{n_i}$ with pairwise coprime n_i :

- Write $x = n_1t + a_1$.
- Substitute into next congruence, solve for t .
- Repeat. Solution unique mod $N = \prod n_i$.

Infinitely many solutions: $x = x_0 + Nk$ for $k \in \mathbb{Z}$.

Show rings not isomorphic

Find invariant that differs:

- Characteristic
- Size
- Number of units
- Number of zero divisors

Ex: $\mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/12\mathbb{Z}$ vs $\mathbb{Z}/108\mathbb{Z}$:

char = $\text{lcm}(9, 12) = 36 \neq 108$. Not isomorphic.

|units| = $\varphi(9)\varphi(12) = 24 \neq 36 = \varphi(108)$.

When does hom exist?

Group hom $C_m \rightarrow C_n$:

Nontrivial exists iff $\gcd(m, n) > 1$.

Ex: $C_{15} \rightarrow C_{30}$: True since $\gcd(15, 30) = 15 > 1$.

Ring hom $\mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ (unital):

Exists iff $n \mid m$ (target divides source).

Ex: $\mathbb{Z}/15\mathbb{Z} \rightarrow \mathbb{Z}/30\mathbb{Z}$: False since $30 \nmid 15$.

Why? $\phi(1) = 1 \Rightarrow \phi(m) = m \cdot 1 = 0$ in target $\Rightarrow n \mid m$.

Quick Reference

Partitions: $p(1) = 1, p(2) = 2, p(3) = 3, p(4) = 5, p(5) = 7, p(6) = 11$.

Totients: $\varphi(1) = 1, \varphi(2) = 1, \varphi(3) = 2, \varphi(4) = 2, \varphi(5) = 4, \varphi(6) = 2, \varphi(8) = 4, \varphi(9) = 6, \varphi(10) = 4, \varphi(12) = 4$.

Common mistakes:

- $\mathbb{Z}[X]$ is NOT a PID: $(2, X)$ not principal.
- $\mathbb{Z}/n\mathbb{Z}$ field iff n prime.
- Ring hom must send $1 \mapsto 1$.